

Lecture 9: The Chain Rule for Multivariable Functions

Background & Motivation

In Calc I, the chain rule handled compositions like $f(g(x))$. Now our functions have multiple intermediate variables, each depending on one or more parameters. The multivariate chain rule keeps track of all the paths from input to output—and in matrix form, it becomes the Jacobian.

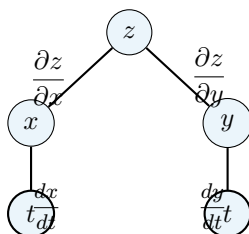
1. Chain Rule: One Parameter (Case 1)

Theorem 1: Chain Rule – Case 1: $z = f(x, y)$, $x = x(t)$, $y = y(t)$

If $z = f(x, y)$ where x and y are both functions of a single parameter t :

$$\frac{dz}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt}$$

– Each intermediate variable contributes one term: (partial w.r.t. that variable) $\times$ (derivative of that variable w.r.t. t)



Tree diagram for Case 1. z depends on x and y ; each of x, y depends on t . To find $\frac{dz}{dt}$, walk every path from z down to t , multiply the derivatives along each path, and sum over all paths.

Example 1: $z = x^2y$ with $x = t^2$, $y = \sin t$

Find $\frac{dz}{dt}$.

1. Compute partials and derivatives:

$$\frac{\partial z}{\partial x} = 2xy = 2t^2 \sin t, \quad \frac{dx}{dt} = 2t$$

$$\frac{\partial z}{\partial y} = x^2 = t^4, \quad \frac{dy}{dt} = \cos t$$

2. Apply chain rule:

$$\frac{dz}{dt} = (2t^2 \sin t)(2t) + (t^4)(\cos t)$$

3. Result: $\frac{dz}{dt} = 4t^3 \sin t + t^4 \cos t$

2. Chain Rule: Multiple Parameters (Case 2)

Theorem 2: Chain Rule – Case 2: $z = f(x, y)$, $x = x(u, v)$, $y = y(u, v)$

$$\frac{\partial z}{\partial u} = \frac{\partial f}{\partial x} \frac{\partial x}{\partial u} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial u}$$

$$\frac{\partial z}{\partial v} = \frac{\partial f}{\partial x} \frac{\partial x}{\partial v} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial v}$$

Matrix Form of the Chain Rule

$$\begin{pmatrix} \frac{\partial z}{\partial u} \\ \frac{\partial z}{\partial v} \end{pmatrix} = \begin{pmatrix} \frac{\partial x}{\partial u} & \frac{\partial y}{\partial u} \\ \frac{\partial x}{\partial v} & \frac{\partial y}{\partial v} \end{pmatrix} \begin{pmatrix} \frac{\partial z}{\partial x} \\ \frac{\partial z}{\partial y} \end{pmatrix}$$

– The chain rule in matrix form is the Jacobian matrix

Example 2: $z = \ln(xy)$ with $x = e^u$, $y = \cos v$

Find $\frac{\partial z}{\partial u}$ and $\frac{\partial z}{\partial v}$.

1. Partial derivatives of z :

$$\frac{\partial z}{\partial x} = \frac{1}{x}, \quad \frac{\partial z}{\partial y} = \frac{1}{y}$$

2. Partial derivatives of x, y :

$$\frac{\partial x}{\partial u} = e^u, \quad \frac{\partial y}{\partial u} = 0, \quad \frac{\partial x}{\partial v} = 0, \quad \frac{\partial y}{\partial v} = -\sin v$$

3. Apply chain rule:

$$\frac{\partial z}{\partial u} = \frac{1}{x} \cdot e^u + \frac{1}{y} \cdot 0 = \frac{e^u}{e^u} = \boxed{1}$$

$$\frac{\partial z}{\partial v} = \frac{1}{x} \cdot 0 + \frac{1}{y} \cdot (-\sin v) = \frac{-\sin v}{\cos v} = \boxed{-\tan v}$$

3. Implicit Differentiation

Implicit Differentiation via Partial Derivatives

If $F(x, y) = 0$ defines y implicitly as a function of x :

$$\frac{dy}{dx} = -\frac{F_x}{F_y} \quad (\text{provided } F_y \neq 0)$$

If $F(x, y, z) = 0$ defines z implicitly as a function of x and y :

$$\frac{\partial z}{\partial x} = -\frac{F_x}{F_z}, \quad \frac{\partial z}{\partial y} = -\frac{F_y}{F_z}$$

– *Why it works:* Differentiate $F(x, y(x)) = 0$ with respect to x using the chain rule: $F_x + F_y \frac{dy}{dx} = 0$, then solve

Example 3: Implicit Differentiation: $x^3 + y^3 = 6xy$

Find $\frac{dy}{dx}$.

1. **Set up:** Let $F(x, y) = x^3 + y^3 - 6xy$

2. **Compute partials:**

$$F_x = 3x^2 - 6y, \quad F_y = 3y^2 - 6x$$

3. **Apply formula:**

$$\frac{dy}{dx} = -\frac{3x^2 - 6y}{3y^2 - 6x} = \boxed{\frac{2y - x^2}{y^2 - 2x}}$$

4. Total Differentials

Definition 1: Total Differential

The **total differential** of $f(x, y)$:

$$df = f_x dx + f_y dy$$

– This is the best linear approximation: $\Delta f \approx df$ for small $dx = \Delta x$, $dy = \Delta y$

For $f(x, y, z)$: $df = f_x dx + f_y dy + f_z dz$

Example 4: Temperature Estimation with Differentials

The temperature in a room is modeled by $T(x, y) = 100 - x^2 - 2y^2$ (degrees). A sensor moves from $(1, 2)$ to $(1.05, 2.1)$. Estimate the change in temperature.

1. **Compute partials at $(1, 2)$:**

$$T_x = -2x = -2, \quad T_y = -4y = -8$$

2. **Apply differential:**

$$dT = T_x dx + T_y dy = (-2)(0.05) + (-8)(0.1) = -0.1 - 0.8$$

3. **Result:** $dT = \boxed{-0.9 \text{ degrees}}$ (temperature drops by $\approx 0.9^\circ$)

Practice Problems

Problem 1. If $z = x^2 + y^2$, $x = t$, $y = e^t$, find $\frac{dz}{dt}$.

Answer: $\frac{dz}{dt} = 2x \cdot 1 + 2y \cdot e^t = 2t + 2e^{2t}$

Problem 2. Let $w = xy + yz + xz$, $x = u + v$, $y = u - v$, $z = uv$. Find $\frac{\partial w}{\partial u}$.

$$w_x = y + z, w_y = x + z, w_z = x + y. \quad x_u = 1, y_u = 1, z_u = v.$$

$$\frac{\partial w}{\partial u} = (y + z)(1) + (x + z)(1) + (x + y)(v)$$

$$= (u - v + uv) + (u + v + uv) + v(2u) = 2u + 2uv + 2uv = \boxed{2u(1 + 2v)}$$

Problem 3. Find $\frac{dy}{dx}$ for the circle $x^2 + y^2 = 25$ using implicit differentiation with partials.

$$F(x, y) = x^2 + y^2 - 25. \quad F_x = 2x, F_y = 2y.$$

$$\frac{dy}{dx} = -\frac{2x}{2y} = \boxed{-\frac{x}{y}}$$

Problem 4. Use differentials to approximate $\sqrt{(3.02)^2 + (3.97)^2}$.

$$f(x, y) = \sqrt{x^2 + y^2}. \text{ At } (3, 4): f = 5, f_x = x/f = 3/5, f_y = y/f = 4/5.$$

$$df = \frac{3}{5}(0.02) + \frac{4}{5}(-0.03) = 0.012 - 0.024 = -0.012$$

$$\sqrt{(3.02)^2 + (3.97)^2} \approx 5 - 0.012 = \boxed{4.988}$$

Problem 5. If $z = f(x - y, xy)$ where f is differentiable, express $x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y}$ in terms of f_1 and f_2 (partials of f w.r.t. its first and second arguments).

$$\text{Let } u = x - y, v = xy. \text{ Then } z_x = f_1 \cdot 1 + f_2 \cdot y \text{ and } z_y = f_1 \cdot (-1) + f_2 \cdot x.$$

$$xz_x + yz_y = x(f_1 + yf_2) + y(-f_1 + xf_2) = (x - y)f_1 + 2xyf_2 = \boxed{uf_1 + 2vf_2}$$